

AKTU Hub - Comprehensive Project Specification Blueprint

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Project Scope: Centralized Academic Platform for Dr. A.P.J. Abdul Kalam Technical University (AKTU) Students

1. Executive Summary & Problem Definition

Students affiliated with AKTU frequently encounter significant friction when attempting to access foundational academic resources. Core components such as lecture notes, official syllabi, previous year question papers (PYQs), and examination results are fragmented across disparate external domains, unorganized Google Drive directories, and an often-unresponsive official portal. This fragmentation causes unnecessary searching time and lowers overall student productivity.

AKTU Hub is engineered as a highly responsive, centralized web application designed to consolidate these scattered resources into a unified interface. By offering an intuitive, mobile-first design, a structured data architecture, and a personalized user dashboard, AKTU Hub eliminates navigational inefficiencies, providing students with immediate access to their specific academic materials.

2. Core Architectural Features

To successfully drive consistent engagement, AKTU Hub must transform from a collection of hyperlinks into a functional ecosystem split into four major components:

A. Multi-Filtered Material Engine

- **Unified Search Architecture:** Direct indexing of academic content allowing queries by both *Subject Name* and exact *Subject Code*. This accounts for frequent university curriculum updates (e.g., transitions from the KAS series to the BAS series).
- **Resource Categorization:** Content partitioned clearly by Semester, Subject Code, and Resource Type (Syllabus, Hand-written Notes, and Year-wise PYQs).

B. Smart CGPA / SGPA Calculator

- **Pre-Loaded Credit Matrices:** Instead of requiring manual input for every course credit value, the tool pre-loads templates directly aligned with specific engineering tracks (e.g., Computer Science & Engineering, Electronics & Communication, Mechanical Engineering) for any selected semester.
- **Real-Time Computations:** Instantaneous credit-weighted calculations based on letter

grades, eliminating manual computation errors.

C. Unified OneView Mirror Dashboard

- **Historical Academic Tracking:** A client-side or database-backed tracking grid allowing users to manually log and archive their semester marks.
- **Progress Visualizations:** Light, clean JavaScript charting (e.g., simple bar or line graphs) plotting academic trajectory over multiple semesters to display dynamic growth metrics.

D. Crowdsourced Elective Directory

- **Peer Review Insights:** A platform feature where senior students can provide qualitative and quantitative ratings regarding departmental electives, scoring them on pass reliability, syllabus difficulty, and resource availability.

3. User Interface (UI) & User Experience (UX) Blueprint

Maximizing usability during stressful academic windows requires adherence to high-standard design guardrails:

A. Visual Identity & Theme

- **Modern Dark Aesthetic:** A premium, low-contrast dark mode to actively reduce visual fatigue during late-night exam preparation. Use deep charcoal surfaces (e.g., #121212) combined with crisp, high-readability typography (Arial or system default sans-serif).
- **Accent Strategy:** Muted blues or clean grays dedicated strictly to functional elements like interactive buttons, table headers, or progress bars.

B. Mobile-First Layout Responsiveness

- **Collapsible Data Grids:** Dense layouts (such as credit matrices, internal mark structures, or exam schedules) must collapse into bite-sized, touch-friendly UI cards when detected on smaller viewports.
- **Persistent Navigation:** A simple thumb-accessible bottom navigation bar on mobile layouts to ensure quick switching between the Dashboard, Notes Engine, and Calculator.

C. Frictionless Access Management

- **Open Resources:** General resource consumption (browsing syllabi, downloading notes, reading PYQs) must remain completely un-gated. No sign-up flow should block a student seeking quick info.
- **Targeted Authentication:** Restrict user login actions purely to personalized features, such as preserving dashboard metrics, storing individual marks records, or submitting crowdsourced notes.

4. Data Architecture & Scalability Matrix

As the application evolves past static asset delivery, migrating towards an organized relational data layout ensures stability and speed.

A. Relational Database Schema Suggestion

Table Name	Primary / Foreign Keys	Core Attributes	Architectural Purpose
Users	user_id (PK)	email, full_name, branch_id, current_semester	Manages user identities and dashboard profiles.
Branches	branch_id (PK)	branch_name, department_code	Maps engineering streams (e.g., CSE, ECE, ME).
Subjects	subject_id (PK), branch_id (FK)	subject_name, subject_code, credits, semester	Stores exact syllabus courses and associated credits.
Resources	resource_id (PK), subject_id (FK)	title, resource_type, storage_url, uploaded_by	Indexes locations for PDFs, notes documents, and PYQs.

B. Asset Storage Decoupling Strategy

- **Lean Metadata:** Databases must only contain lightweight text fields, metadata strings, and object URL paths.
- **Cloud Bucket Isolation:** Heavy document payloads (such as 10MB scanned exam papers or textbook PDFs) should be hosted externally on scalable storage solutions (e.g., Supabase Storage or AWS S3 buckets) to preserve lightning-fast query execution.

5. Student Retention & Growth Strategy

Building a valuable portal is only half the battle; capturing recurring user traffic requires capitalizing on specific academic cycles:

A. Search Engine Optimization (SEO) Engineering

- **High-Intent Keywords:** Students run highly specific searches right before sessionals. Structure your pages to match exact patterns, such as: "*AKTU [Subject Name] [Subject Code] Notes PDF*" or "*AKTU [Semester] CSE PYQ 2025*".
- **Semantic Structure:** Utilize exact page titles and meta-descriptions matching these academic search phrases to ensure search engines rank AKTU Hub above outdated online forums.

B. Gamified Crowdsourcing Engine

- **Peer Contributions:** Allow verified users to upload high-quality handwritten study notes directly through their dashboard.
- **Public Validation:** Build a simple upvoting or download counter system. High-performing documents earn their authors visibility on a community "Top Contributors" leaderboard, building peer recognition and ensuring high-quality, up-to-date data.